

BARCLAYS INTERVIEW EXPERIENCE FOR INTERNSHIP

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INTRODUCTION :

Barclays first sent a gform for the registration of candidates for their online exams. And out of more than 400 students applied they shortlisted merely 180 students for online assessment based on their cgpa(8.3 and above). As at that time I had a cgpa of around 8.5 I was shortlisted for the OA which had 20 mcq's(technical mainly output based) and 5 SQL queries(difficulty varies from 3 questions easy and 2 questions medium) main constrain in this OA is time management as all of these should be completed within 45 minutes. And after the OA after a week they sent the shortlisted list for the next round in which they kept on adding names at the end and the final shortlisted list contained about 47 candidates(Even though some of the candidates attended only 24 questions out of 25 and cannot answer the last SQL also got shortlisted as 47 is a large amount in 180 candidates). I have answered all the 25 questions and 5 SQL queries passed all the test cases so I am somewhat sure that I would go into the next round and fortunately I was shortlisted.

ROUND-1(TECHNICAL ROUND) :

Barclays conducted 2 rounds of interview out of which the first round is the technical round. There were around 10 panels for the interview and each panel consisted a single interviewer. Out of those 10 panels 4 panels were difficult 3 panels were normal and 3 panels were very easy like they asked HR-Level questions at the technical round. As always my luck I went into one among the 4 difficult panels and I was listed as the third candidate to attend the interview at my panel. The thing is that when I attended for Appian interview you can communicate with others who have completed their interview to get their insights but Barclays strict doesn't allow this and they send all the completed candidates to a different area. So we were blank about what is happening in the interview room but I read my seniors interview experience as they all said it is the easiest interview process they ever attended

so I am confident that it would be simple, but hold on folks it is clearly the opposite that went for my panel. After the previous candidate before me completed his interview I was called next and I went in and the first thing he did his read my resume thoroughly pin-pointing at critical tech skills. I was asked to introduce about myself as usual the interviewers don't listen to your intro instead at that time they would read what and all you have added in your resume and google it to ask potential questions from it. So after this he asked about my strengths and skills I explained it to him. And he directly went onto something that I haven't even specified in my resume 'matplotlib'. He given me a set of data like it is the shopping list of region specific customers in certain months and he wanted me to plot that in any way I know using any language. So I went on with "charts.io" package from node js and explained how would I pull the data from api's and plot it and I also explained about matplotlib in python and told him that I was never used to visualization in python and I would go with node js. He was happy with my answer and moved on to the next question where he saw I used the word JWT Authentication in my project and asked me how does it works. I explained the concept behind it and he went further deeper and asked how does the passwords are getting hashed and I explained him the usage of bcrypt he was still not happy and said some certain protocol numbers and asked me which protocol I was using I had no answer for those specific protocols and I said only I know the tool to hash and I am currently reading its protocols. Then he moved on to cryptography and asked how do you encode your data and how would you retrieve the data at the other end without dampening of data. I told him about the concepts of middleware and basic cryptography techniques to mask the data which remembered when studying computational thinking in first semester. Then he asked how would you make your product available for every servers. I explained him the process of horizontal scaling and he was fine with it. Then he moved onto dbms and he wrote about 4 tables and asked me to retrieve certain data from the tables. It involved using groupby, join, subquery and having clause finally sorted by using order by clause. After that he asked me about row indexing methods that I know. I explained him about clustered and non-cluster indexing. Then he again wrote a table and asked me to manipulate the table based on the data being inserted and I asked him whether I can proceed with procedure he green flagged me and I wrote a PL/SQL procedure which generally modifies a date when the region column is changed you can do it by using cursors in procedures. Then he moved on to dsa and asked me

whether I know doubly linked list. I said yes and he asked me to write the code for deletion in the doubly linked list at a specific node. I wrote a pseudocode and diagrammatically explained him what is happening. Then he moved on to networks and I told him that we are currently pursuing networks and OS in this semester but it doesn't stop him from asking questions. He asked me to explain about the protocols I know. I explained him TCP and UDP and he asked a follow up question of explain the seven segments in the networks layer. At that time I was able to remember only 4 and I explained it to him he said that was fine. Then he moved onto OS and he gave me ABC.dat file and asked me to write a command that remove all the junk character from the file(characters other than alphabets) from the file and store it in a new file. At that time, I only know about grep, sed and awk I managed to write a simple program in awk that does this work. Then he asked about cybersecurity while working in bank kind of systems how would you ensure security. He asked me to explain about the different types of ai's that I know. I explained him about ai, gen ai, open ai, agentic ai and so on. He followed up with give examples for each of those. Then he asked me whether I have knowledge on github as I didn't host any project in github he asked the reasons for it. After he asked me about various github commands like clone, merge, commit, pull, fork and after I explained he then asked me whether I know about BOLT AI . This is the point where I said I completely have no clue. Then he stopped the interview(in between I think he is waiting for this word I don't know in the complete interview). He asked me whether I have any questions for him. I asked him about the tech stacks I would work at Barclays. After one hour of long gruelling interview it came to an end and he said "it has been nice talking to you man" but definitely not the same from me. I came out with a sigh of relief and a bit of confidence that I have done reasonably well.

KEY POINTS :

- Be well prepared with dbms concepts and sql querying (Oracle is preferable).
- Go through commonly asked dsa questions of Barclays.
- Know a little bit about LLD(Low Level Design) and revise OOPs concept thoroughly.
- And last but not the least you should have some luck by your hand as only the 4 panels out of 10 panels the interview was tough the other guys easily escaped with simple dsa and sql querying and the interview

went for only 20 to 25 minutes for some candidates. Only in your bad day you will get on to me like I attended a one hour long interview.

ROUND – 2 (Technical + HR) :

After the first round was completed for all the candidates they shortlisted 25 students out of the 47 initially qualified students. I was fortunate to be one among them. And I noticed that everyone from my panel was eliminated except me. Then I came to realize they have relatively selected candidates from every panel. When asked what about the second round they said it will be project deep dives and situation based hr questions. In the second round I was announced as the first candidate and without a breathing time they asked me to get into the room. This was so quick such that I actually forgot my intro and somehow managed when he asked tell me about yourself. Then he told me that we will divide it as 50 to 50 technical and hr divided. He was the sweetest person I have ever talked in 5 rounds of interview(3 in Appian and 2 in Barclays, don't ask me about Appian they told me silliest of reasons to reject me after going through all the 3 rounds). The interviewer asked shall we proceed with the tech stacks you know. I said yes and he asked me some basic questions about what is node js, why react, for backend development do you use node js or react and why, why react is not suitable for backend development. He asked me about the best feature I like using node js. I answered him npm packet manager. He asked me why and what are the things you have done using npm. I explained him about creating live servers and all. Then he asked how components are transferred from one state another state for example he draw a 3-tier client server architecture and asked to explain me what is going behind the abstraction and what are the tools you know used for this purpose. I explained the concepts of POSTMAN and REST api's and detailed him about CRUD operations. I also explained him about middleware and horizontal scaling. Then he moved on to MONGO DB and asked questions like why you prefer mongo db, is mongo db indexed or not indexed, if not indexed then give me the command that makes it indexed according to user, how do you create a collection, what is a document in mongo db, mongo db better than RDBMS. Then he dived on to some dbms questions. Why normalization and denormalization, is cartesian product good or bad, if bad what is the alternative, give real life examples of worst cases of cartesian product, what are the aggregate functions I know, explain me about some window functions,

when to use surrogate key, is surrogate key good or bad, types of indexing methods, difference between rank and dense rank, when will you go for rank in real life scenarios, how to optimize queries, drawbacks of RDBMS. Then he went on to OOPs concepts where he asked questions which were never topic specific he would tell some scenarios and I want to identify which phenomenon of OOP here is taking place. For example, he asked me a class just has some declarations and no variables and asked me what is the reason for creating this pseudo class. The answer is runtime polymorphism as the methods will be overridden in the child classes like this no question was specific and it was seriously testing your knowledge of OOPs in real life scenarios. Then he asked me about what is the favorite feature you like in c++, I explained him about friend classes. And he jokingly asked me whether I would use this while working in Barclays company. I strictly said no as Barclays contains sensitive information of persons in their database. He was happy with this much questions for technical purpose and with a smile he said I have grilled you enough for a internship purpose and we can move on to scenario based question. By this time snacks arrived and we took a bite and continued the interview(in between he was very heartious to share it with me). He asked you have written code for 6 months for a project and one of your friendly has deleted the code accidentally or deliberately that you don't know and you have presentation tomorrow and your talent exposure chance is based on this what will you do. I answered this in a more saving my friend nature way but he reminded me that when a boat is going to sink you have to save yourself first before lending your hand to others. Then he asked me about situations if you have mis-communication between your team and it is your fault and how would you resolve it. The he asked me to explain a situation where I have followed any one of the Barclays principle(RISES). I explained him about a story that tells about integrity. And that's time it has already been nearly an hour he asked me whether I have any questions for him. And winded up the interview by saying looking forward to see you. That's when I got a hope I would get selected and it happened fortunately so.

KEY POINTS :

- The second round mostly contains project deep dives so study thoroughly about your projects and the things you have used in your project.

- They look for confidence in hr round so be confident about your answer as everyone have different answers no answer is correct it's all about delivering your thoughts.
- Finally try something out of the box to impress the interviewer by including many real life scenarios as he is the one to select you for the intern.

ALL THE BEST